

ECMT676 Econometrics II

Tentative Solutions to Problem Set 1

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1 Analytical Problems

1. **(2.4)** Since $E(u|\mathbf{x}, v)$ is a linear function of $(\mathbf{x}, v)$, $L(u|\mathbf{x}, v)$ is also a linear function of $(\mathbf{x}, v)$ by PROPERTY LP.1. Denote $L(u|\mathbf{x}, v) = \mathbf{x}\beta + \rho_1 v$ and $r_1 \equiv \mathbf{x} - L(\mathbf{x}|v)$. By PROPERTY LP.7, $L(u|r_1) = r_1\beta$. Since $\mathbf{Cov}(v, \mathbf{x}) = 0$, we know that $r_1 = \mathbf{x}$. Moreover, since $\mathbf{Cov}(u, \mathbf{x}) = 0$, we can get that $\beta = 0$. Thus, $E(u|\mathbf{x}, v) = E(u|v) = \rho_1 v$.
2. **(2.6)**
 - (a) $L(q|1, \mathbf{x}) = L(q^* + e|1, \mathbf{x}) = L(q^*|1, \mathbf{x}) + L(e|1, \mathbf{x}) = L(q^*|1, \mathbf{x})$. The last equality comes from $E(e)$ and $E(\mathbf{x}'e) = \mathbf{0}^1$.
 - (b) $r \equiv q - L(q|1, \mathbf{x}) = q^* + e - L(q|1, \mathbf{x}) = L(q|1, \mathbf{x}) + r^* + e - L(q|1, \mathbf{x}) = r^* + e$.
3. **(3.6)** $\frac{\hat{\theta}_N - \theta}{N^{-c}} = \frac{\sqrt{N}(\hat{\theta}_N - \theta)}{\sqrt{N}N^{-c}} = \frac{1}{N^{1/2-c}} \sqrt{N}(\hat{\theta}_N - \theta) = o(1)O_p(1) = o_p(1)$ and thus $\hat{\theta}_N - \theta = o_p(N^{-c})$.
4. **(5.12)** Following the hint, we know that $E(\mathbf{z}'\mathbf{x}) = E(\mathbf{z}'\mathbf{z})\Pi$. Note that

$$\Pi' = \begin{bmatrix} \mathbf{I}_{K-1} & \mathbf{0}_{(K-1) \times M} \\ \delta & \theta \end{bmatrix}$$

where $\delta = [\delta_1, \dots, \delta_{K-1}]$ and $\theta = [\theta_1, \dots, \theta_M]$. Thus, by the way of instruction of Π , we know that $\text{rank}(E(\mathbf{z}'\mathbf{x})) = K$ if and only if at least one θ_j is different from zero.

¹Recall that $L(e|1, \mathbf{x}) = \alpha + \mathbf{x}\beta$ where $\beta = [\text{Var}(\mathbf{x})]^{-1}\text{Cov}(\mathbf{x}, e)$ and $\alpha = E(e) - E(\mathbf{x})\beta$.

2 Empirical Problems

1. (5.6)

(a)	Var.	coef	stde	t-stat	p-val	lb of CI	ub of CI
	const	4.7394	0.0715	66.2588	0.0000	4.5991	4.8796
	educ	0.0747	0.0035	21.3510	0.0000	0.0678	0.0816
	exper	0.0848	0.0066	12.8063	0.0000	0.0718	0.0978
	exper ²	-0.0023	0.0003	-7.2232	0.0000	-0.0029	-0.0017
	black	-0.1990	0.0182	-10.9058	0.0000	-0.2348	-0.1632
	south	-0.1480	0.0260	-5.6950	0.0000	-0.1989	-0.0970
	smsa	0.1364	0.0201	6.7851	0.0000	0.0970	0.1758
	reg661	-0.1186	0.0388	-3.0536	0.0023	-0.1947	-0.0424
	reg662	-0.0222	0.0283	-0.7857	0.4321	-0.0776	0.0332
	reg663	0.0260	0.0274	0.9491	0.3427	-0.0277	0.0796
	reg664	-0.0635	0.0357	-1.7795	0.0753	-0.1335	0.0065
	reg665	0.0095	0.0361	0.2618	0.7935	-0.0614	0.0803
	reg666	0.0219	0.0401	0.5473	0.5842	-0.0567	0.1006
	reg667	-0.0006	0.0394	-0.0150	0.9881	-0.0778	0.0766
	reg668	-0.1750	0.0463	-3.7766	0.0002	-0.2659	-0.0841
	smsa66	0.0262	0.0194	1.3493	0.1773	-0.0119	0.0644
(b)	Var.	coef	stde	t-stat	p-val	lb of CI	ub of CI
	const	16.8485	0.2111	79.8046	0.0000	16.4346	17.2625
	exper	-0.4125	0.0337	-12.2415	0.0000	-0.4786	-0.3465
	exper ²	0.0009	0.0017	0.5263	0.5987	-0.0024	0.0041
	black	-0.9355	0.0937	-9.9806	0.0000	-1.1193	-0.7517
	south	-0.0516	0.1354	-0.3811	0.7032	-0.3172	0.2139
	smsa	0.4022	0.1048	3.8372	0.0001	0.1967	0.6077
	reg661	-0.2103	0.2025	-1.0386	0.2991	-0.6072	0.1867
	reg662	-0.2889	0.1473	-1.9608	0.0500	-0.5778	-0.0000
	reg663	-0.2382	0.1426	-1.6701	0.0950	-0.5179	0.0415
	reg664	-0.0931	0.1860	-0.5005	0.6167	-0.4578	0.2716
	reg665	-0.4829	0.1882	-2.5660	0.0103	-0.8519	-0.1139
	reg666	-0.5131	0.2096	-2.4475	0.0144	-0.9241	-0.1020
	reg667	-0.4271	0.2056	-2.0771	0.0379	-0.8303	-0.0239
	reg668	0.3136	0.2417	1.2977	0.1945	-0.1602	0.7875
	smsa66	0.0255	0.1058	0.2409	0.8096	-0.1819	0.2329
	nearc4	0.3199	0.0879	3.6408	0.0003	0.1476	0.4922
(c)	Var.	coef	stde	t-stat	p-val	lb of CI	ub of CI
	const	3.7740	0.9325	4.0473	0.0002	1.9456	5.6023
	educ	0.1315	0.0548	2.3989	0.8954	0.0240	0.2390
	exper	0.1083	0.0236	4.5886	0.9138	0.0620	0.1545
	exper ²	-0.0023	0.0003	-7.0201	0.9981	-0.0030	-0.0017
	black	-0.1468	0.0538	-2.7304	0.8833	-0.2522	-0.0414
	south	-0.1447	0.0272	-5.3165	0.8850	-0.1980	-0.0913
	smsa	0.1118	0.0316	3.5407	0.9110	0.0499	0.1737
	reg661	-0.1078	0.0417	-2.5853	0.9142	-0.1896	-0.0260
	reg662	-0.0070	0.0328	-0.2147	0.9944	-0.0714	0.0573

reg663		0.0404	0.0317	1.2760	0.9677	-0.0217	0.1026
reg664		-0.0579	0.0375	-1.5442	0.9538	-0.1315	0.0156
reg665		0.0385	0.0468	0.8215	0.9693	-0.0533	0.1302
reg666		0.0551	0.0525	1.0489	0.9561	-0.0479	0.1581
reg667		0.0268	0.0487	0.5495	0.9787	-0.0687	0.1222
reg668		-0.1909	0.0506	-3.7743	0.8486	-0.2901	-0.0917
smsa66		0.0185	0.0216	0.8599	0.9852	-0.0237	0.0608

(d) i. Var. | coef stde t-stat p-val lb of CI ub of CI

const		16.7731	0.2163	77.5281	0.0000	16.3489	17.1973
exper		-0.4123	0.0337	-12.2373	0.0000	-0.4784	-0.3462
exper^2		0.0008	0.0017	0.5139	0.6074	-0.0024	0.0041
black		-0.9452	0.0939	-10.0650	0.0000	-1.1293	-0.7610
south		-0.0419	0.1355	-0.3092	0.7572	-0.3077	0.2238
smsa		0.4014	0.1048	3.8304	0.0001	0.1959	0.6068
reg661		-0.1688	0.2041	-0.8270	0.4083	-0.5689	0.2314
reg662		-0.2690	0.1478	-1.8198	0.0689	-0.5589	0.0208
reg663		-0.1902	0.1458	-1.3049	0.1920	-0.4760	0.0956
reg664		-0.0377	0.1892	-0.1994	0.8420	-0.4086	0.3332
reg665		-0.4371	0.1903	-2.2967	0.0217	-0.8103	-0.0639
reg666		-0.5022	0.2097	-2.3951	0.0167	-0.9134	-0.0911
reg667		-0.3775	0.2079	-1.8157	0.0695	-0.7852	0.0302
reg668		0.3820	0.2454	1.5566	0.1197	-0.0992	0.8632
smsa66		0.0001	0.1069	0.0007	0.9994	-0.2096	0.2098
nearc2		0.1230	0.0774	1.5886	0.1123	-0.0288	0.2748
nearc4		0.3206	0.0878	3.6495	0.0003	0.1483	0.4928

ii. Var. | coef stde t-stat p-val lb of CI ub of CI

const		3.3397	0.8922	3.7434	0.0008	1.5904	5.0890
educ		0.1571	0.0524	2.9951	0.8752	0.0542	0.2599
exper		0.1188	0.0227	5.2237	0.9054	0.0742	0.1634
exper^2		-0.0024	0.0003	-6.7990	0.9981	-0.0030	-0.0017
black		-0.1233	0.0520	-2.3702	0.9019	-0.2253	-0.0213
south		-0.1432	0.0284	-5.0476	0.8861	-0.1988	-0.0876
smsa		0.1008	0.0314	3.2051	0.9198	0.0391	0.1624
reg661		-0.1030	0.0433	-2.3778	0.9180	-0.1879	-0.0181
reg662		-0.0002	0.0337	-0.0068	0.9998	-0.0663	0.0659
reg663		0.0470	0.0326	1.4420	0.9626	-0.0169	0.1108
reg664		-0.0554	0.0391	-1.4179	0.9558	-0.1320	0.0212
reg665		0.0515	0.0474	1.0856	0.9589	-0.0415	0.1445
reg666		0.0700	0.0532	1.3166	0.9442	-0.0342	0.1742
reg667		0.0391	0.0496	0.7872	0.9688	-0.0582	0.1363
reg668		-0.1980	0.0524	-3.7797	0.8430	-0.3008	-0.0953
smsa66		0.0151	0.0223	0.6762	0.9880	-0.0286	0.0587

(e) Var. | coef stde t-stat p-val lb of CI ub of CI

const		100.6106	0.6275	160.3470	0.0000	99.3801	101.8412
nearc4		2.5962	0.7455	3.4825	0.0005	1.1342	4.0582

(f) Var. | coef stde t-stat p-val lb of CI ub of CI

const		99.3847	0.7017	141.6416	0.0000	98.0089	100.7605
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nearc4	0.8681	0.8217	1.0565	0.2909	-0.7431	2.4792
smsa66	1.3545	0.8028	1.6873	0.0917	-0.2196	2.9286
reg661	4.7681	1.5468	3.0825	0.0021	1.7352	7.8010
reg662	5.8081	0.9018	6.4409	0.0000	4.0400	7.5762
reg669	1.8447	1.1517	1.6017	0.1094	-0.4136	4.1029